

Analysis of the health of Montgomery County, Maryland

Based on Social Determinants of Health from the U.S. Dept. of Health and Human Services

Mychala Walker

Introduction

For this paper, I intend to accomplish two things: 1) determine the overall health of Montgomery County, Maryland and 2) compare the health status of the different Census tracts of Montgomery County, Maryland. I employ the U.S. Department of Health and Human Services' five (5) social determinant groups of health. The social determinant groups are economic stability, education access and quality, health care access and quality, neighborhood and built environment, & social and community context. Within these groups are a number of variables which will be explained in great detail below. To compare the tracts, I will create eight (8) k-means clusters of the Census tracts and analyze the clusters, both demographically and based on the social determinants of health factors. The total within sum of square method was used to determine the optimal number of clusters for this model.

Data

For this study, I combine multiple datasets from two sources: 1) the 2022 U.S. American Community Survey (ACS) and 2) Montgomery County, Maryland Open Data Portal. The combined data set had 232 observations where every observation is a census tract. Census tracts are small, relatively permanent statistical subdivisions of the County from the U.S. Census Bureau. According to the the U.S. Census Bureau, each area contains approximately 4,000 people of relatively homogeneous characteristics with respect to population, economic status and living conditions. Census tracts provide a stable set of geographic units for the presentation of decennial census data over the decades and are only occasionally split to reflect population growth. The model estimated that eight (8) clusters was optimal using the the total within sum of squares method.

The dataset had 42 variables that represent both demographic and social determinants of health based on standards set by U.S. Human and Health Services. According to the U.S.

Human and Health Services, social determinants of health (SDOH) are the conditions in the environments where people are born, live, learn, work, play, worship, and age that affect a wide range of health, functioning, and quality-of-life outcomes and risks. Social determinants of health also contribute to wide health disparities and inequities. Variables on employment, number of grocery stores, housing insecurity, and poverty are used to represent economic stability. Variables on early childhood development and poverty, enrollment in higher education, high school graduation are used to represent education access and quality. Variables on the number of HHS facilities are used to represent health care access and quality. Variables on number of grocery stores, amount of crimes, and neighborhood walk score are used to represent neighborhood and built environment. There were no variables used to represent social and community context. All of the variables are numeric variables except for the GEOID variable. There is also a geometry variable for graphing purposes.

Analysis

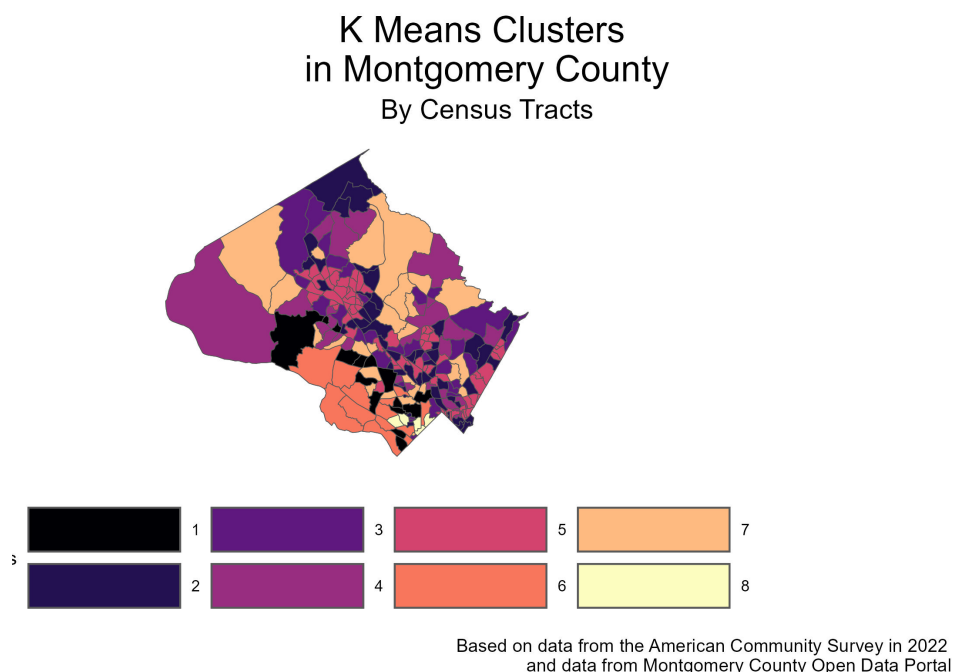


Figure 1

As we can see from the above graphs, there are a few clusters that seem to be sectioned to one part of the county. For example, clusters one (1), six (6), and eight (8) seem to only be in the bottom part of the county. Further, cluster five (5) seems to be portioned mostly in the center part of the county. The mean demographics of the clusters themselves are interesting.

A tibble: 8 x 7

	cluster	mean_pop	mean_whipop	mean_blkpop	mean_hispop	mean_income	mean_age
*	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	4441.	69.6	4.63	7.55	287558.	45.3
2	2	4814.	44.5	19.4	22.4	138586.	39.0
3	3	4743.	47.4	17.2	18.0	169925.	40.9
4	4	4617.	55.7	13.0	11.4	197021.	43.4
5	5	4461.	31.1	28.7	30.0	101300.	37.8
6	6	4725.	69.8	5.12	6.78	357641.	48.1
7	7	4262.	63.0	9.15	9.48	224252.	45.0
8	8	3302.	82.3	2	4.7	492099.	49.6

Cluster 8 has a significantly high white population, high mean income, high mean life expectancy and low poverty. On the other hand, cluster 5 has a significantly high minority population, high unemployment, high poverty, and low life expectancy. Clusters 1, 6, and 7 are on the high end for white population and low minority population. Unemployment rate is similar across the clusters, with the exception of clusters 5 and 8, the unemployment rate are around 3 and 4. The mean income across the clusters range from \$101,000 and \$493,000. The average age is also pretty similar across all the clusters with a range of 37 and 49 years old. Interestingly, cluster 5 is the “youngest” cluster whereas cluster 8 is the “oldest”.

```
# A tibble: 8 x 11
  cluster mean_travelwrk own_rent_rto mean_unigrad mean_schoolenroll
*   <int>         <dbl>         <dbl>         <dbl>         <dbl>
1     1         31.1         6.26         2542.         1246
2     2         33.1         1.78         1893.         1163.
3     3         32.6         2.80         1916.         1246.
4     4         34.2         5.16         2221.         1196.
5     5         33.0         0.751        1292.         1090.
6     6         30.0         9.18         2874.         1259.
7     7         33.2         7.78         2158.         1133.
8     8         29.3        11.8         2148.          871.
# i 6 more variables: mean_noinsure <dbl>, mean_unemploy <dbl>,
#   mean_violent <dbl>, mean_poverty <dbl>, mean_lifeexpectancy <dbl>,
#   mean_grocery <dbl>
```

As for the social determinants of health, we see a similar trend. Cluster 8 is an all around wealthy and healthy cluster. Their own to rent ratio is significantly higher than than the other clusters. Their population with no insurance is low and their number of violent crimes is low. As for cluster 5, we see the complete opposite. Their no-insurance population is the highest amongst all of the clusters. Their violent crime rate is also extremely high compared to the other clusters. Based on these findings and the standards set by HHS, this cluster would be deemed unhealthy. Interestingly, amongst all of the clusters, the university graduate population

rate is similar. The own to rent ratio between the clusters are also starkingly different. With the exception of cluster eight (8), child age school enrollment is similar.

Conclusion

Based on these findings and the standards set by HHS, comparatively, clusters 1, 6, 7, and 8 are the healthiest clusters in Montgomery county. On the other side, clusters 2, 3, 4, and 5 are the least healthy. This means that representatives and local officials from representing clusters 2, 3, 4, and 5 to consider programming that may improve social conditions, with the goal of improving health outcomes.